



Objective Exercise - I

1. If R is a relation from a finite set A having m elements to a finite set B having n elements, then the number of relations from A to B is-
(1) 2^{mn} (2) $2^{mn} - 1$ (3) $2mn$ (4) m^n
2. In the set $A = \{1, 2, 3, 4, 5\}$, a relation R is defined by $R = \{(x, y) \mid x, y \in A \text{ and } x < y\}$. Then R is-
(1) Reflexive (2) Symmetric (3) Transitive (4) None of these
3. For real numbers x and y , we write $x R y \Leftrightarrow x - y + \sqrt{2}$ is an irrational number. Then the relation R is.
(1) Reflexive (2) Symmetric (3) Transitive (4) None of these
4. Let $X = \{1, 2, 3, 4\}$ and $Y = \{1, 3, 5, 7, 9\}$. Which of the following is relations from X to Y -
(1) $R_1 = \{(x, y) \mid y = 2 + x, x \in X, y \in Y\}$
(2) $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$
(3) $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)\}$
(4) $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$
5. Let L denote the set of all straight lines in a plane. Let a relation R be defined by $\alpha R \beta \Leftrightarrow \alpha \perp \beta, \alpha, \beta \in L$. Then R is.
(1) Reflexive (2) Symmetric (3) Transitive (4) None of these
6. Let R be a relation defined in the set of real numbers by $a R b \Leftrightarrow 1 + ab > 0$. Then R is-
(1) Equivalence relation (2) Transitive
(3) Symmetric (4) Anti-symmetric
7. Which one of the following relations on R is equivalence relation-
(1) $x R_1 y \Leftrightarrow |x| = |y|$ (2) $x R_2 y \Leftrightarrow x \geq y$
(3) $x R_3 y \Leftrightarrow x \mid y$ (4) $x R_4 y \Leftrightarrow x < y$
8. Two points P and Q in a plane are related if $OP = OQ$, where O is a fixed point. This relation is-
(1) Reflexive but not symmetric (2) Symmetric but not transitive
(3) An equivalence relation (4) none of these
9. The relation R defined in $A = \{1, 2, 3\}$ by $a R b$ if $|a^2 - b^2| \leq 5$. Which of the following is false-
(1) $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (1, 2), (2, 3), (3, 2)\}$
(2) $R^{-1} = R$
(3) Domain of $R = \{1, 2, 3\}$
(4) Range of $R = \{5\}$
10. Let a relation R in the set N of natural numbers be defined as $(x, y) \in R$ if and only if $x^2 - 4xy + 3y^2 = 0$ for all $x, y \in N$. The relation R is-
(1) Reflexive (2) Symmetric
(3) Transitive (4) An Equivalence relation



11. Let $A = \{2, 3, 4, 5\}$ and let $R = \{(2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (3, 5), (5, 3)\}$ be a relation in A . Then R is-
- (1) Reflexive and transitive (2) Reflexive and symmetric
(3) Reflexive and antisymmetric (4) none of these
12. If $A = \{2, 3\}$ and $B = \{1, 2\}$, then $A \times B$ is equal to-
- (1) $\{(2, 1), (2, 2), (3, 1), (3, 2)\}$ (2) $\{(1, 2), (1, 3), (2, 2), (2, 3)\}$
(3) $\{(2, 1), (3, 2)\}$ (4) $\{(1, 2), (2, 3)\}$
13. Let R be a relation over the set $N \times N$ and it is defined by $(a, b) R(c, d) \Rightarrow a + d = b + c$. Then R is.
- (1) Reflexive only (2) Symmetric only
(3) Transitive only (4) An equivalence relation
14. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b) R(c, d)$ if $ad(b + c) = bc(a + d)$, then R is-
- (1) Symmetric only (2) Reflexive only
(3) Transitive only (4) An equivalence relation
15. If $A = \{1, 2, 3\}$, $B = \{1, 4, 6, 9\}$ and R is a relation from A to B defined by ' x is greater than y '. Then range of R is-
- (1) $\{1, 4, 6, 9\}$ (2) $\{4, 6, 9\}$ (3) $\{1\}$ (4) none of these
16. Let L be the set of all straight lines in the Euclidean plane. Two lines ℓ_1 and ℓ_2 are said to be related by the relation R if ℓ_1 is parallel to ℓ_2 . Then the relation R is-
- (1) Reflexive (2) Symmetric (3) Transitive (4) Equivalence
17. A and B are two sets having 3 and 4 elements respectively and having 2 elements in common. The number of relations which can be defined from A to B is -
- (1) 2^5 (2) $2^{10} - 1$ (3) $2^{12} - 1$ (4) 2^{12}
18. For $n, m \in N$, $n \mid m$ means that n is a factor of m , the relation $\mid$ is -
- (1) reflexive and symmetric
(2) transitive and symmetric
(3) reflexive, transitive and symmetric
(4) reflexive, transitive and not symmetric
19. Let $R = \{(x, y) : x, y \in A, x + y = 5\}$ where $A = \{1, 2, 3, 4, 5\}$ then
- (1) R is not reflexive, symmetric and not transitive
(2) R is an equivalence relation
(3) R is reflexive, symmetric but not transitive
(4) R is not reflexive, not symmetric but transitive
20. Let R be relation on a set A such that $R = R^{-1}$ then R is-
- (1) reflexive (2) symmetric (3) transitive (4) none of these



- 21.** Let $x, y \in \mathbb{Z}$ and suppose that a relation R on $\mathbb{Z}$ is defined by $x R y$ if and only if $x \leq y$ then
(1) R is partial order relation (2) R is an equivalence relation
(3) R is reflexive and symmetric (4) R is symmetric and transitive
- 22.** Let R be a relation from a set A to a set B , then
(1) $R = A \cup B$ (2) $R = A \cap B$ (3) $R \subseteq A \times B$ (4) $R \subseteq B \times A$
- 23.** Given the relation $R = \{(1, 2), (2, 3)\}$ on the set $A = \{1, 2, 3\}$, then minimum number of ordered pairs which when added to R make it an equivalence relation
(1) 5 (2) 6 (3) 7 (4) 8
- 24.** Let $P = \{(x, y) \mid x^2 + y^2 = 1, x, y \in \mathbb{R}\}$ Then P is -
(1) reflexive (2) symmetric (3) transitive (4) anti-symmetric
- 25.** Let X be a family of sets and R be a relation on X defined by ' A is disjoint from B '. Then R is -
(1) reflexive (2) symmetric (3) anti-symmetric (4) transitive
- 26.** In order that a relation R defined in a non-empty set A is an equivalence relation, it is sufficient that R
(1) is reflexive (2) is symmetric
(3) is transitive (4) possesses all the above three properties
- 27.** If R is an equivalence relation in a set A , then R^{-1} is -
(1) reflexive but not symmetric (2) symmetric but not transitive
(3) an equivalence relation (4) none of these
- 28.** Let $A = \{p, q, r\}$. Which of the following is an equivalence relation in A ?
(1) $R_1 = \{(p, q), (q, r), (p, r), (p, p)\}$ (2) $R_2 = \{(r, q), (r, p), (r, r), (q, q)\}$
(3) $R_3 = \{(p, p), (q, q), (r, r), (p, q)\}$ (4) none of these

ANSWER KEY

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|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|
| 1. | (1) | 2. | (3) | 3. | (1) | 4. | (1) | 5. | (2) | 6. | (3) | 7. | (1) |
| 8. | (3) | 9. | (4) | 10. | (1) | 11. | (2) | 12. | (1) | 13. | (4) | 14. | (4) |
| 15. | (3) | 16. | (4) | 17. | (4) | 18. | (4) | 19. | (1) | 20. | (2) | 21. | (1) |
| 22. | (3) | 23. | (3) | 24. | (2) | 25. | (2) | 26. | (4) | 27. | (3) | 28. | (4) |

JEE-Main (Previous Year Questions)

1. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$, The relation R is -
 (1) transitive (2) not symmetric [AIEEE-2004]
 (3) reflexive (4) a function

2. Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be relation on the set $A = \{3, 6, 9, 12\}$. The relation is- [AIEEE-2005]
 (1) reflexive and transitive only
 (2) reflexive only
 (3) an equivalence relation
 (4) reflexive and symmetric only

3. Let W denote the words in the English dictionary. Define the relation R by :
 $R = \{(x, y) \in W \times W \mid \text{the words } x \text{ and } y \text{ have at least one letter in common}\}$. Then R is- [AIEEE-2006]
 (1) reflexive, symmetric and not transitive
 (2) reflexive, symmetric and transitive
 (3) reflexive, not symmetric and transitive
 (4) not reflexive, symmetric and transitive

4. Consider the following relations :-
 $R = \{(x, y) \mid x, y \text{ are real numbers and } x = wy \text{ for some rational number } w\}$;
 $S = \left\{ \left(\frac{m}{n}, \frac{p}{q} \right) \mid m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn \right\}$. Then : [AIEEE-2010]
 (1) R is an equivalence relation but S is not an equivalence relation
 (2) Neither R nor S is an equivalence relation
 (3) S is an equivalence relation but R is not an equivalence relation
 (4) R and S both are equivalence relations

5. Let R be the set of real numbers. [AIEEE-2011]
Statement-1 : $A = \{(x, y) \in R \times R : y - x \text{ is an integer}\}$ is an equivalence relation on R .
Statement-2 : $B = \{(x, y) \in R \times R : x = \alpha y \text{ for some rational number } \alpha\}$ is an equivalence relation on R .
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.



6. Consider the following two binary relations on the set $A = \{a, b, c\}$:
 $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ and
 $R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$.
Then
(1) R_1 is not symmetric but it is transitive
(2) both R_1 and R_2 are transitive
(3) R_2 is symmetric but is not transitive
(4) both R_1 and R_2 are not symmetric **[JEE(Main)-2018]**
7. Let N denote the set of all natural numbers. Define two binary relations on N as
 $R_1 = \{(x, y) \in N \times N : 2x + y = 10\}$ and
 $R_2 = \{(x, y) \in N \times N : x + 2y = 10\}$. Then :
(1) Both R_1 and R_2 are symmetric relations
(2) Range of R_1 is $\{2, 4, 8\}$
(3) Both R_1 and R_2 are transitive relations
(4) Range of R_2 is $\{1, 2, 3, 4\}$ **[JEE(Main)-2018]**
8. Let R_1 and R_2 be two relations defined as follows :
 $R_1 = \{(a, b) \in R^2 : a^2 + b^2 \in Q\}$ and
 $R_2 = \{(a, b) \in R^2 : a^2 + b^2 \notin Q\}$, where Q is the set of all rational numbers. Then :
(1) R_1 is transitive but R_2 is not transitive. (2) R_1 and R_2 are both transitive.
(3) R_2 is transitive but R_1 is not transitive. (4) Neither R_1 nor R_2 is transitive. **[JEE(Main)-2020]**
9. Let $X = \{n \in N : 1 \leq n \leq 50\}$. If $A = \{n \in X : n \text{ is a multiple of } 2\}$ and $B = \{n \in X : n \text{ is a multiple of } 7\}$, then the number of elements in the smallest subset of X containing both A and B is..... **[JEE(Main)-2020]**
10. Define a relation R over a class of $n \times n$ real matrices A and B as “ ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$ ”. Then which of the following is true? **[JEE(Main)-2020]**
(1) R is symmetric, transitive but not reflexive,
(2) R is reflexive, symmetric but not transitive
(3) R is an equivalence relation
(4) R is reflexive, transitive but not symmetric
11. Let N be the set of natural numbers and a relation R on N be defined by
 $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is: **[JEE(Main)-2020]**
(1) symmetric but neither reflexive nor transitive
(2) reflexive but neither symmetric nor transitive
(3) reflexive and symmetric, but not transitive
(4) an equivalence relation



12. Let R_1 and R_2 be relations on the set $\{1, 2, \dots, 50\}$ such that $R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and $R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$. Then, the number of elements in $R_1 - R_2$ is ____
[JEE(Main)-2020]
13. Let R_1 and R_2 be two relations defined on R by $a R_1 b \Leftrightarrow ab \geq 0$ and $a R_2 b \Leftrightarrow a \geq b$, then
[JEE(Main)-2020]
(A) R_1 is an equivalence relation but not R_2
(B) R_2 is an equivalence relation but not R_1
(C) both R_1 and R_2 are equivalence relations
(D) neither R_1 nor R_2 is an equivalence relation
14. For $\alpha \in N$, consider a relation R on N given by $R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$. The relation R is an equivalence relation if and only if:
[JEE(Main)-2020]
(A) $\alpha = 14$
(B) α is a multiple of 4
(C) 4 is the remainder when α is divided by 10
(D) 4 is the remainder when α is divided by 7
15. Let R be a relation from the set $\{1, 2, 3, \dots, 60\}$ to itself such that $R = \{(a, b) : b = pq, \text{ where } p, q \geq 3 \text{ are prime numbers}\}$. Then, the number of elements in R is:
[JEE(Main)-2020]
(A) 600 (B) 660 (C) 540 (D) 720
16. Which of the following is not correct for relation R on the set of real numbers?
[JEE(Main)-2021]
(1) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric
(2) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive
(3) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric
(4) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric
17. Let N be the set of natural numbers and a relation
[JEE(Main)-2021]
 R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$
Then the relation R is
(1) Symmetric but neither reflexive nor transitive
(2) reflexive but neither symmetric nor transitive
(3) reflexive but symmetric, but not transitive
(4) an equivalence relation



18. Let $A = \{2, 3, 4, 5, \dots, 30\}$ and ' $\simeq$ ' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to [JEE(Main)-2021]
(1) 5 (2) 6 (3) 8 (4) 7
19. Let R_1 and R_2 be two relations defined on $\mathbb{R}$ by $a R_1 b \Leftrightarrow ab \geq 0$ and $a R_2 b \Leftrightarrow a \geq b$, then [JEE(Main)-2022]
(1) R_1 is an equivalence relation but not R_2
(2) R_2 is an equivalence relation but not R_1
(3) both R_1 and R_2 are equivalence relations
(4) neither R_1 nor R_2 is an equivalence relation
20. For $\alpha \in \mathbb{N}$, consider a relation R on $\mathbb{N}$ given by $R = \{(x, y) : 3x + \alpha y \text{ is multiple of } 7\}$. The relation R is an equivalence relation if and only if: [JEE(Main)-2022]
(1) $\alpha = 14$
(2) α is multiple of 4
(3) 4 is the remainder when α is divided by 10
(4) 4 is the remainder when α is divided by 7
21. Let R be a relation from the set $\{1, 2, 3, \dots, 60\}$ to itself such that $R = \{(a, b) : a = pq, \text{ where } p, q \geq 3 \text{ are prime numbers}\}$. Then the number of elements in R is: [JEE(Main)-2022]
(1) 600 (2) 660 (3) 540 (4) 720
22. Let R_1 and R_2 be relations on the set $\{1, 2, \dots, 50\}$ such that $R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and $R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$. Then, the number of elements in $R_1 = R_2$ is _____. [JEE(Main)-2022]
23. Let $R_1 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in \mathbb{N} \times \mathbb{N} : |a - b| \neq 13\}$, Then on $\mathbb{N}$: [JEE(Main)-2022]
(1) Both R_1 and R_2 are equivalence relations
(2) Neither R_1 nor R_2 is an equivalence relation
(3) R_1 is an equivalence relation but R_2 is not
(4) R_2 is an equivalence relation but R_1 is not
24. Let a set $A = A_1 \cup A_2 \cup \dots \cup A_k$ where $A_i \cap A_j = \emptyset$ for $i \neq j, 1 \leq i, j \leq k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i \text{ if and only if } x \in A_i, 1 \leq i \leq k\}$. Then, R is : [JEE(Main)-2022]
(1) reflexive, symmetric but not transitive
(2) reflexive, transitive but not symmetric
(3) reflexive but not symmetric and transitive
(4) an equivalence relation



ANSWER KEY

1.	(2)	2.	(1)	3.	(1)	4.	(3)	5.	(1)	6.	(3)	7.	(4)
8.	(4)	9.	29.00	10.	(3)	11.	(2)	12.	8	13.	(D)	14.	(D)
15.	(B)	16.	(2)	17.	(1)	18.	(4)	19.	(4)	20.	(4)	21.	(2)
22.	(8)	23.	(2)	24.	(4)								

